

- Excelerate

Data Visualization Associate Early Internship Final Report

From: Team 24

To Our Sponsor: Excelerate Team

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WEEK 4 Deliverables
From: Team 24

- Excelerate

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Project Overview:

- This Project provides an in-depth examination of three interconnected datasets—**ApplicantData**, **OutreachData**, and **CampaignData**—from the Illinois Institute of Technology's admissions pipeline for AY2024. The exploratory data analysis (PRELIMINARY DATA EXPLORATION) uncovered key patterns, irregularities, and trends such as duplicate applicant records, inconsistent phone number formats, missing or invalid country entries, imbalanced caller workloads, and inadequately recorded outreach remarks. Campaign analysis showed a stronger focus on post-admission engagement compared to pre-admission recruitment, with most outreach efforts concentrated in a limited number of campaigns.
- Data cleaning procedures standardized formats, removed duplicates, and categorized outcomes more effectively, preparing the datasets for reliable integration into PostgreSQL for structured analysis. The results highlighted both advantages—such as a large applicant pool and well-organized campaign schedules—and weaknesses, including high rates of unsuccessful calls, insufficient remark documentation, and data inconsistencies.

- The project concludes with recommendations aimed at improving data quality, balancing operational workloads, optimizing campaign strategies, and incorporating predictive analytics to strengthen applicant engagement and increase enrollment conversions in future admission cycles.

Exploratory Data Analysis:

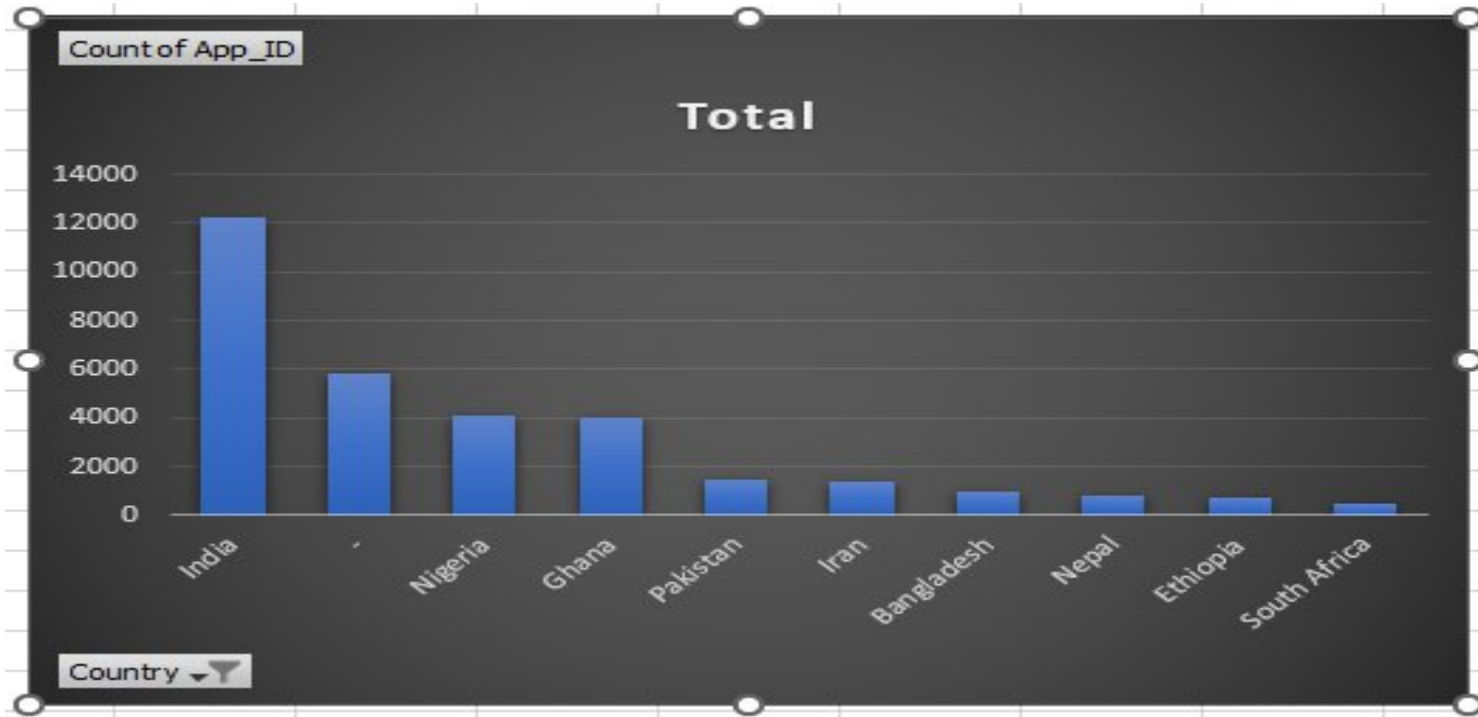
- The goal to gain an initial insight into a dataset by examining its structure, main features, and underlying patterns.
- It is also essential for spotting anomalies, inconsistencies, and issues such as missing or duplicate records.
- **Descriptive Statistics (Uncleaned Data):**
- **Applicant Data Set** (*App_ID*):
 - Total records: **37,882**
 - Unique applicants: **15,416**
 - Duplicates: Many applicants appear multiple times with different phone numbers.
 - Example: App_ID 12345 appears 3+ times.

- ***Countries***
- Unique entries: **856** (suspiciously high → inconsistent naming).
- Top country: **India** with **12,234 records**.
- Other notable ones: Nigeria, Pakistan, China, USA (we'll chart the top 10).

- ***University***
- Only **1 value**: Illinois Institute of Technology.
- Means this dataset is specific to one university's applicants.

- ***Phone Numbers***
- Total records: **37,882**
- Unique numbers: **18,391**
- Numbers vary in format: some 10 digits, some include country codes, others up to 13 digits.

- **Country Distribution**



- The chart shows that **India** contributes the largest number of applicants (~12,000+).
- Nigeria, Ghana, and Pakistan follow as the next major contributors.
- Some countries have very low counts (e.g., Ethiopia, South Africa).

- A few applicant IDs have **large spikes** with more than 1000+ phone numbers linked — this is highly unrealistic and indicates a **data anomaly or duplication issue**.
- Several other numbers repeat around 20–25 times.
- These anomalies will need to be investigated and possibly removed during cleaning.
- Instead of country names, there are **email IDs and other invalid text values** appearing in the “Country” column.
- This indicates that the dataset has **misplaced data**, where email addresses (or other values) were wrongly entered under “Country.”

- **Data Cleaning:**

1. **Duplicate Records**

- **Issue:** Many applicants appeared multiple times with the same App_ID.
- **Solution:** Removed exact duplicate rows.

2. Missing and Invalid Country Values

- **Issue:** The “Country” field contained missing values represented as blanks (“-”) and also invalid entries such as **email addresses** and non-country text.
- **Solution:**
 - Replaced blanks with “Unknown.”
 - Removed or corrected invalid entries (emails, misplaced text).
 - Standardized country names to ensure consistency (e.g., “INDIA”, “India ”, and “Ind.” were standardized to “India”).

3. Repeated Phone Numbers Across Applicants

4. Phone Number Formatting Inconsistencies

- **Issue:** Phone numbers were stored in varying formats (10-digit, 12-digit with country code, leading zeros, etc.).
- **Solution:** Standardized all phone numbers into a consistent format: country code + 10-digit number (where available). Invalid phone numbers (too short/too long) were removed.

- **Exploratory Data Analysis:**

- **Outreach Data Set:**

- **Rows:** 37,881, **Columns:** 8

- **Early Insights:**

- **Reference_ID**

- 15,416 unique applicants (same scale as ApplicantData).
- Many applicants appear multiple times (multiple outreach attempts).

- **Recieved_At**

- 31,236 unique timestamps.
- Shows activity spread over many days (call campaign logs).

- **University**

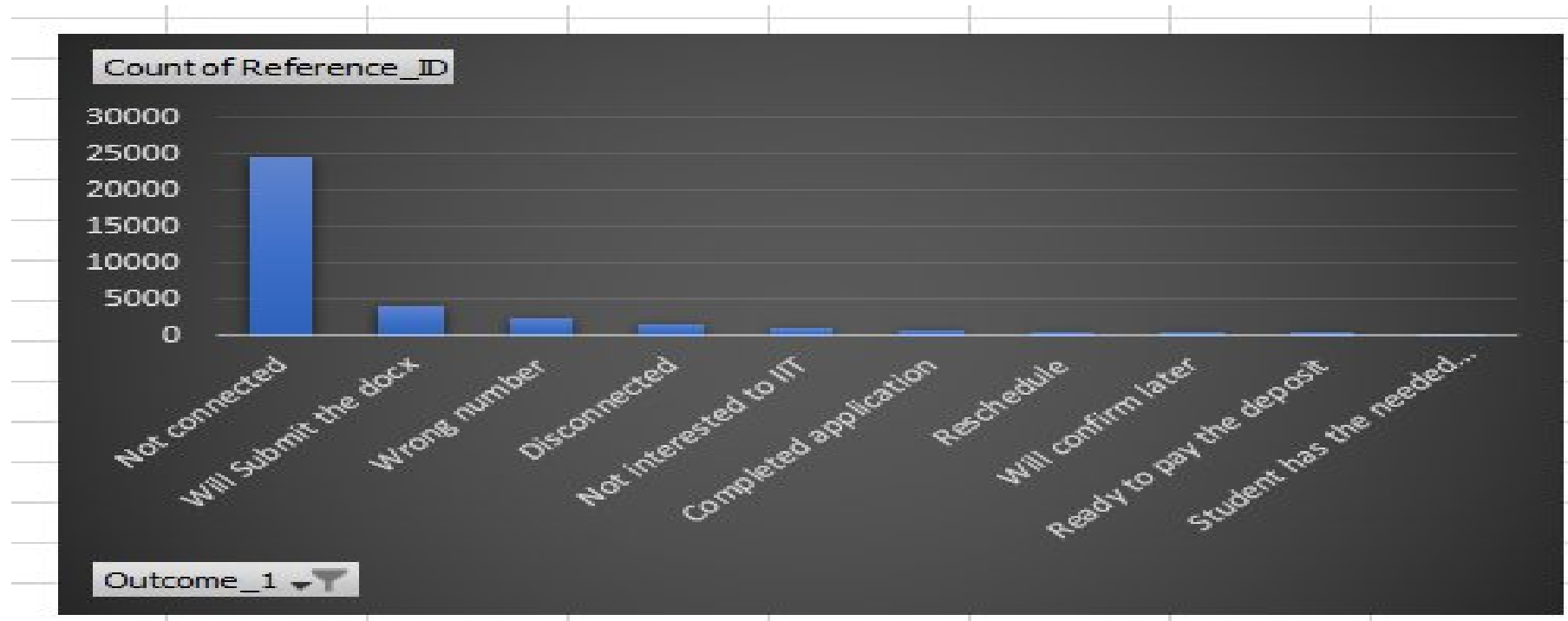
- Only 1 unique value: *Illinois Institute of Technology* (like ApplicantData).

- **Caller_Name**

- 12 unique callers.
- Top caller: **Rudra** (handled 14,712 calls, ~39% of total).
- Suggests uneven workload distribution.

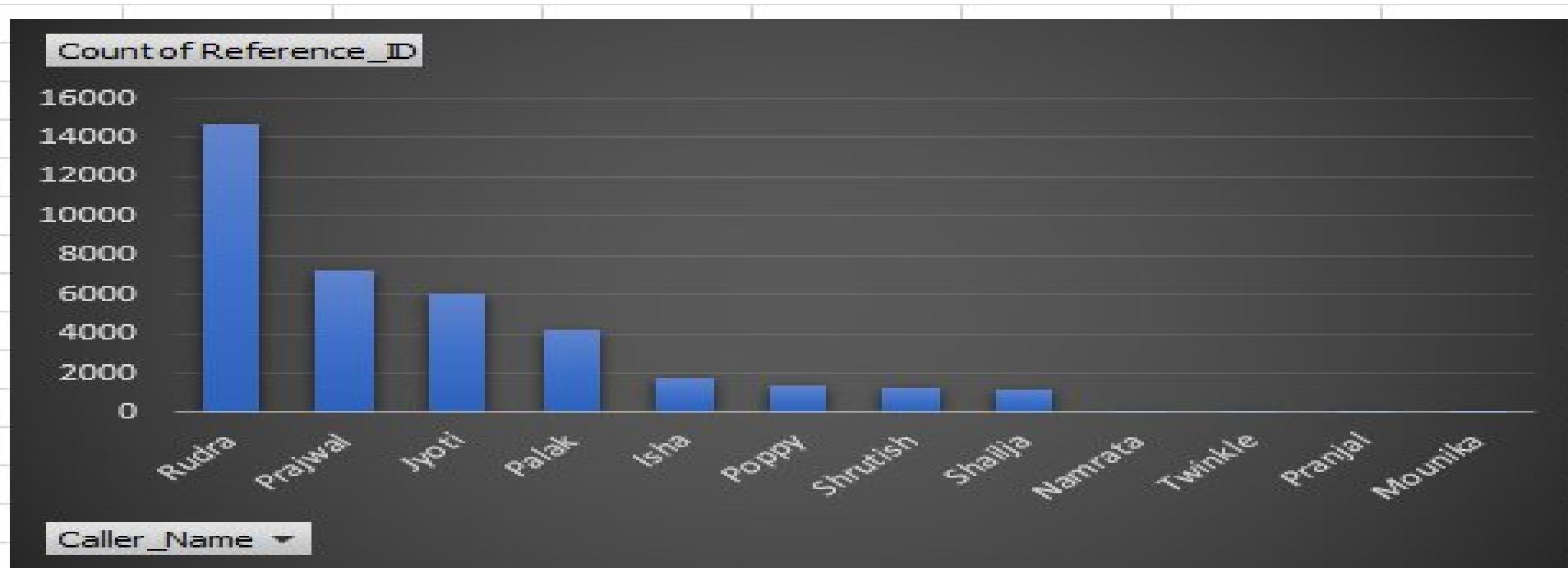
- **Outcome_1**
 - 41 unique outcomes recorded.
 - Most frequent: **“Not connected”** (24,498 times, ~65%).
 - Other outcomes: *Connected, Reschedule, Completed application, Will Submit docx* etc.
 - Shows majority of calls fail to connect.
- **Remark**
 - Very sparse → only **4,077 entries** (~11%).
 - Most frequent remark: **“No answer”** (151 times).
- **Campaign_ID**
 - 23 unique campaigns.
 - Largest: **FA24IP** (9,605 calls).
 - Suggests recurring outreach drives.
- **Escalation_Required**
 - 3 categories (mostly **“No”** → **37,118 times / 98%**).
 - Very few cases needed escalation.

- ***Call Outcomes Distribution***



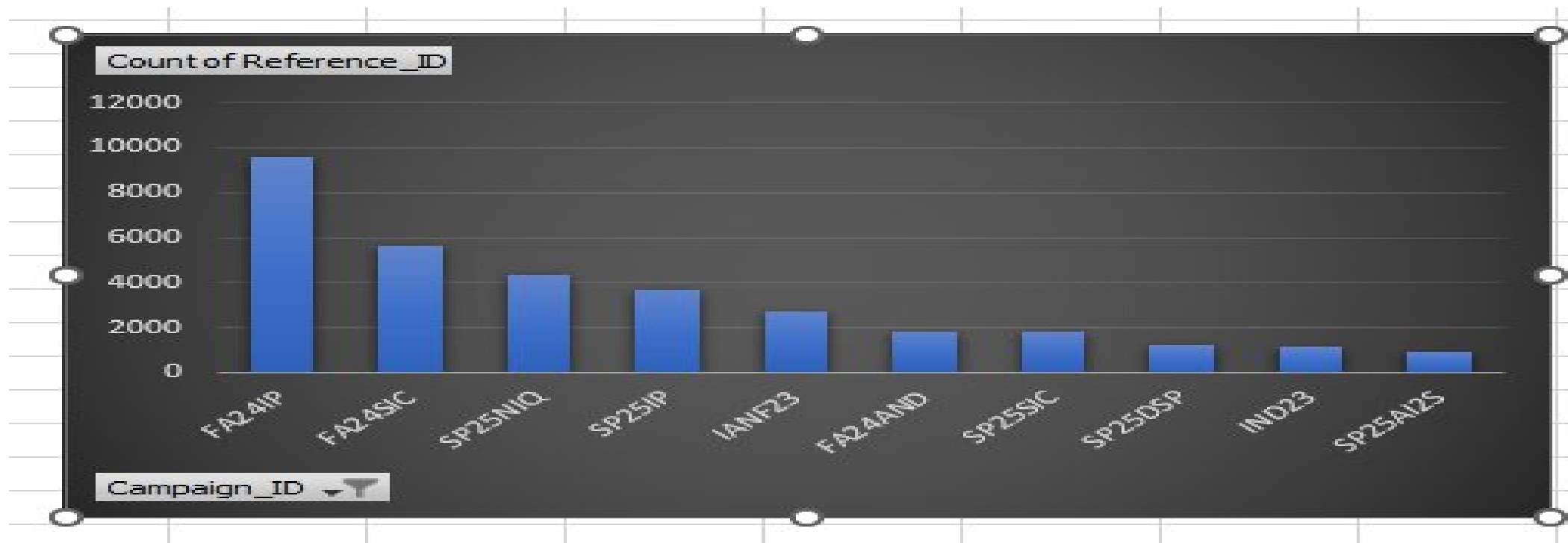
- The dataset shows that the **majority of outreach attempts were unsuccessful**.
- The most common outcome was “**Not connected**” (over 24,000 calls, ~65% of total).

- **Caller Performance**



- **Rudra** handled the largest number of calls (~15,000), nearly **40% of total outreach**.
- **Prajwal** and **Jyoti** follow with ~7,000 and ~6,000 calls respectively.
- Several callers (e.g., Namrata, Twinkle, Pranjal, Mounika) had **very low call counts**.

- ***Campaign Effectiveness***



- The most active campaign was **FA24IP**, with nearly 10,000 calls.
- Other major campaigns include **FA24SIC**, **SP25NIQ**, and **SP25IP**, each ranging from 3,000–6,000 calls.
- Smaller campaigns like **IND23** and **SP25A12S** had fewer than 1,000 calls.

• Data Cleaning:

1. Duplicate Records

- **Issue:** Many applicants (Reference_ID) appear multiple times due to repeated outreach attempts. While some duplication is valid (different call attempts), exact duplicate rows exist.
- **Solution:** Removed exact duplicate rows, but retained multiple unique attempts per applicant for historical accuracy.

2. Missing and Invalid Remarks

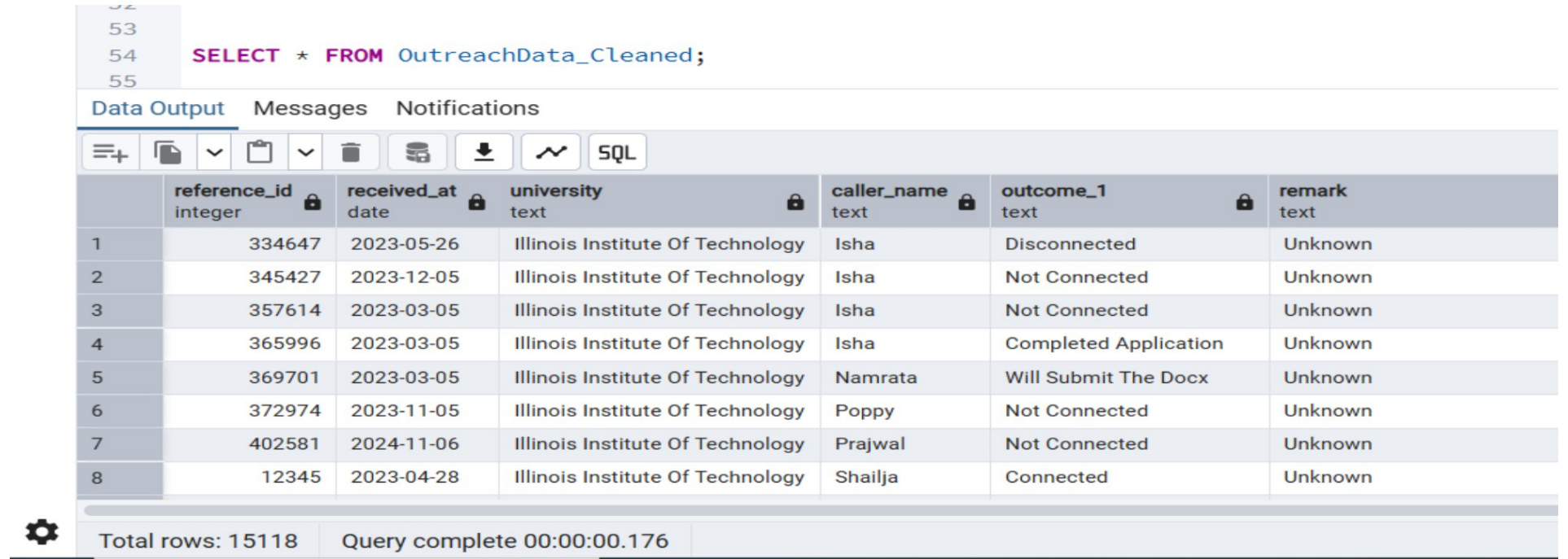
- **Issue:** The Remark column is mostly empty (~89% missing). Where remarks exist, many are inconsistent or incomplete (e.g., “within few days,” “will submit”).
- **Solution:**
 - Replaced blank remarks with a placeholder "No Remark".
 - Standardized recurring remarks (e.g., “No answer,” “Not attending,” “Will submit documents”) into consistent categories.

3. Call Outcomes (Outcome_1)

- **Issue:** Some call outcomes are too detailed or inconsistent (e.g., “Student has the needed...”, “Will confirm later”).
- **Solution:** Grouped outcomes into broader standardized categories.

4. Campaign IDs

- **Issue:** Campaign IDs are consistent (23 unique codes). However, some campaigns are very small and others dominate the dataset.
- **Solution:** Standardized campaign naming (ensured all codes were uppercase and consistent format).



The screenshot displays a database management tool interface. At the top, a SQL query is entered: `SELECT * FROM OutreachData_Cleaned;`. Below the query editor, there are tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is active, showing a table with 8 rows of data. The table has columns: 'reference_id' (integer), 'received_at' (date), 'university' (text), 'caller_name' (text), 'outcome_1' (text), and 'remark' (text). The data rows show various call records from the Illinois Institute Of Technology. At the bottom of the interface, a status bar indicates 'Total rows: 15118' and 'Query complete 00:00:00.176'.

	reference_id integer	received_at date	university text	caller_name text	outcome_1 text	remark text
1	334647	2023-05-26	Illinois Institute Of Technology	Isha	Disconnected	Unknown
2	345427	2023-12-05	Illinois Institute Of Technology	Isha	Not Connected	Unknown
3	357614	2023-03-05	Illinois Institute Of Technology	Isha	Not Connected	Unknown
4	365996	2023-03-05	Illinois Institute Of Technology	Isha	Completed Application	Unknown
5	369701	2023-03-05	Illinois Institute Of Technology	Namrata	Will Submit The Docx	Unknown
6	372974	2023-11-05	Illinois Institute Of Technology	Poppy	Not Connected	Unknown
7	402581	2024-11-06	Illinois Institute Of Technology	Prajwal	Not Connected	Unknown
8	12345	2023-04-28	Illinois Institute Of Technology	Shailja	Connected	Unknown

Total rows: 15118 Query complete 00:00:00.176

- **Exploratory Data Analysis:**

- **Campaign Dataset:**

- **Rows: 23, Columns: 7**

- **Early Insights:**

- ***Unique Campaigns***

- 23 unique campaigns, each with a different ID and Name.
- These match the Campaign_ID field we saw in **OutreachData**.

- ***Category Distribution***

- **14 campaigns are Post Admission, 9 campaigns are Pre Admission.**

- ***Intake***

- All campaigns are tied to **AY2024.**, it Suggests the dataset represents a single admissions cycle.

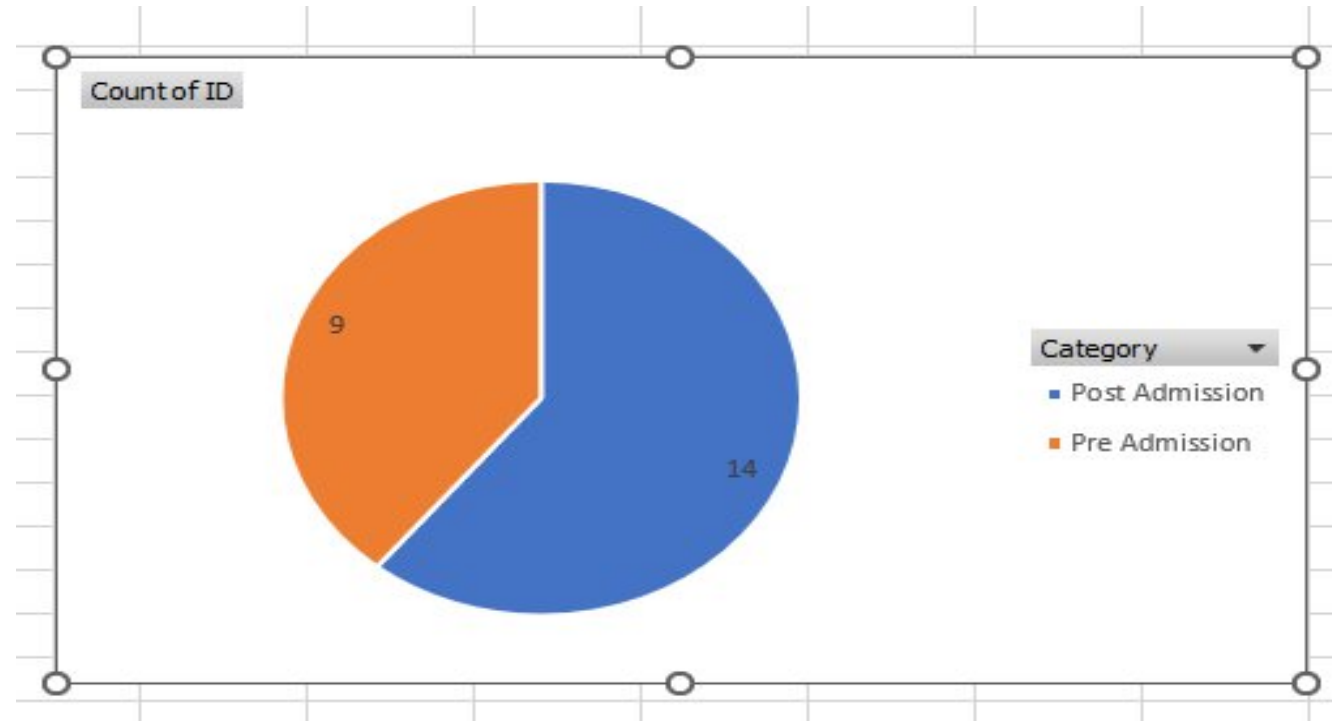
- ***Status***

- Every campaign is marked **Completed**.
- No “In Progress” or “Cancelled,” so this is a closed historical record.

- ***Start Date***

- Campaigns span from **April 2023** to **December 2024**.
- Reflects a rolling campaign strategy: outreach began well before the intake and continued into late 2024.

- ***Campaign Categories***



- The chart shows that out of **23 campaigns**, **14 (61%)** are classified as **Post Admission** and **9 (39%)** as **Pre Admission**.

- **Key Insight:** The campaign strategy is **heavily weighted toward post-admission engagement**, reflecting the institution's priority on **converting admitted students into enrolled students** rather than just generating new applications.

- **Data Cleaning:**

- **1. Campaign IDs (ID)**

- **Issue:** All 23 campaigns had unique IDs, but some contained a mix of upper/lowercase or extra spaces.
- **Solution:** Standardized all campaign IDs to **uppercase and trimmed whitespace** to ensure consistency, especially when joining with OutreachData.

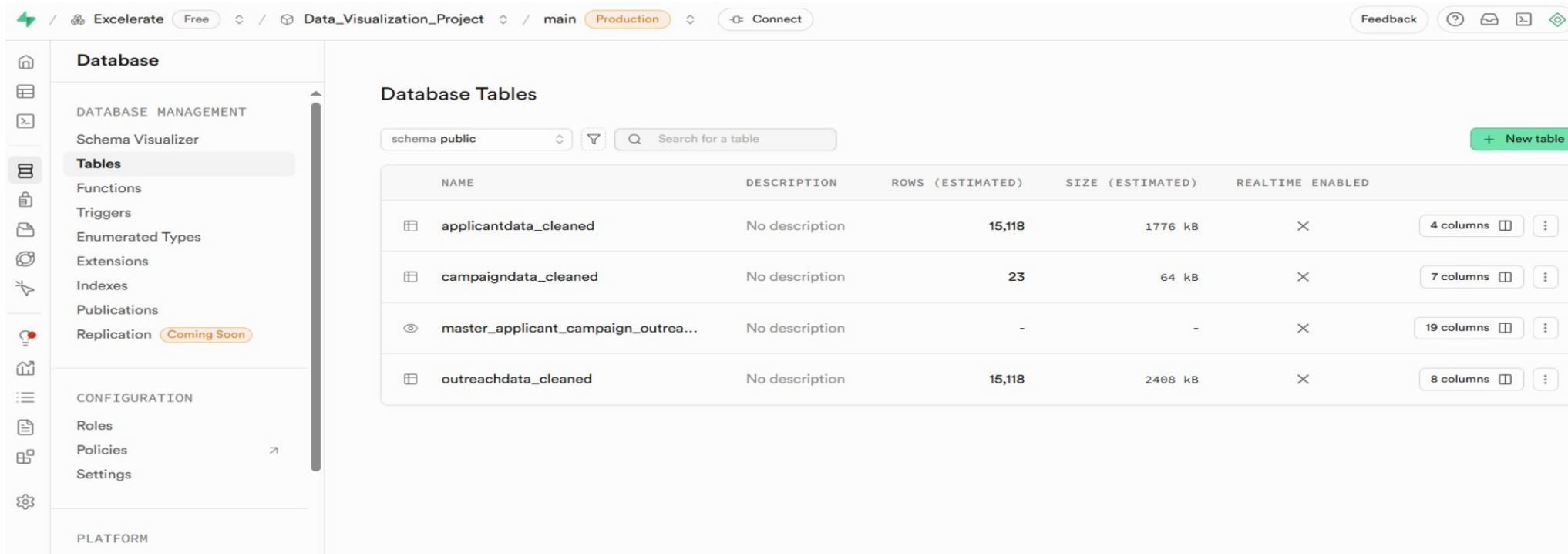
- **2. Campaign Names (Name)**

- **Issue:** Campaign names were descriptive but inconsistent in formatting (e.g., some with hyphens, some with colons).
- **Solution:** Left as-is to preserve descriptive detail, but noted that standardizing naming conventions in future campaigns would improve readability.

- **3. Category**
- **Issue:** Only two categories were present — *Pre Admission* and *Post Admission* — but some records had inconsistent spacing or capitalization.
- **Solution:** Standardized the values to exactly “**Pre Admission**” and “**Post Admission.**”
- **7. Start Date (Start_Date)**
- **Issue:** Dates were stored as text in inconsistent formats (e.g., 3/20/2024 0:00, 9/11/2024 00:00).
- **Solution:** Converted to a consistent **date format (YYYY-MM-DD)** for accurate timeline analysis.
- **Master Table Creation:**
- To integrate all three datasets (**ApplicantData**, **OutreachData**, and **CampaignData**) into a unified structure, a **Master Table** was created in PostgreSQL. This table provides an end-to-end view of each applicant’s journey, linking their application details with outreach attempts and campaign information.

• Data Connection:

- Faced initial challenges connecting via Supabase, resolved after extensive troubleshooting.
- Established Command Prompt queries, uploaded datasets, and optimized indexes.
- Successfully tested static connection through Looker Studio (CSV upload).



The screenshot displays the Supabase Database Management interface. The left sidebar shows the 'Database' section with options like Schema Visualizer, Tables, Functions, Triggers, Enumerated Types, Extensions, Indexes, Publications, and Replication (marked 'Coming Soon'). The main area, titled 'Database Tables', shows a list of tables in the 'public' schema. The table list includes columns for NAME, DESCRIPTION, ROWS (ESTIMATED), SIZE (ESTIMATED), REALTIME ENABLED, and a column for column details. The tables listed are 'applicantdata_cleaned', 'campaigndata_cleaned', 'master_applicant_campaign_outrea...', and 'outreachdata_cleaned'.

NAME	DESCRIPTION	ROWS (ESTIMATED)	SIZE (ESTIMATED)	REALTIME ENABLED	
applicantdata_cleaned	No description	15,118	1776 kB	×	4 columns
campaigndata_cleaned	No description	23	64 kB	×	7 columns
master_applicant_campaign_outrea...	No description	-	-	×	19 columns
outreachdata_cleaned	No description	15,118	2408 kB	×	8 columns

Data Consistency and Validation:

Here we have 3 queries for the 3 main tables, verifying that data types and formats (e.g., dates, numeric fields) are consistent across all tables.

Query Query History

```
1
2
3 SELECT column_name, data_type
4 FROM information_schema.columns
5 WHERE table_schema = 'public'
6     AND table_name = 'applicantdata_cleaned'
7 ORDER BY ordinal_position;
8
9
10
```

Data Output Messages Notifications

SQL

	column_name name	data_type character varying
1	app_id	integer
2	country	text
3	university	text
4	phone_number	text

```
9
10 SELECT column_name, data_type
11 FROM information_schema.columns
12 WHERE table_schema = 'public'
13     AND table_name = 'campaigndata_cleaned'
14 ORDER BY ordinal_position;
```

Data Output Messages Notifications

SQL

	column_name name	data_type character varying
1	id	text
2	name	text
3	category	text
4	intake	text
5	university	text
6	status	text
7	start_date	date

```

14
15 SELECT column_name, data_type
16 FROM information_schema.columns
17 WHERE table_schema = 'public'
18     AND table_name   = 'outreachdata_cleaned'
19 ORDER BY ordinal_position;
20

```

Data Output Messages Notifications



	column_name name	data_type character varying
1	reference_id	integer
2	received_at	date
3	university	text
4	caller_name	text
5	outcome_1	text
6	remark	text
7	campaign_id	text
8	escalation_required	text

Data Consistency and Validation:

Here we have 2 select queries, ensuring that key identifiers like Applicant IDs or Campaign IDs match across related tables (Outreach_Data), all good (0 Rows).

```
24
25 -- Orphaned outreach records (Reference_ID not found in ApplicantData)
26 SELECT o.Reference_ID
27 FROM Outreachdata_Cleaned o
28 LEFT JOIN ApplicantData_Cleaned a
29     ON o.Reference_ID = a.App_ID
30 WHERE a.App_ID IS NULL;
31
32
33
34
35
36
37
```

Data Output Messages Notifications

reference_id
integer

Total rows: 0 Query complete 00:00:00.095

```
32
33 -- Orphaned outreach records (Campaign_ID not found in CampaignData)
34 SELECT o.campaign_id
35 FROM outreachdata_cleaned o
36 LEFT JOIN campaigndata_cleaned c
37     ON o.campaign_id = c.id
38 WHERE c.id IS NULL;
39
```

Data Output Messages Notifications

campaign_id
text

Total rows: 0 Query complete 00:00:00.083

Then we have 2 queries to ensure that the Applicant count and Campaign count in the database match the dashboard, with no discrepancies.

Campaign Count	Applicant Count
23	15,118

```
20
21 SELECT * FROM ApplicantData_Cleaned;
22
23
24
```

	app_id [PK] integer	country text	university text	phone_number text
1	12345	India	Illinois Institute Of Technology	18019011222
2	347397	Nigeria	Illinois Institute Of Technology	9.19183E+11
3	358065	India	Illinois Institute Of Technology	9.19963E+11
4	351333	India	Illinois Institute Of Technology	13173701409
5	346435	India	Illinois Institute Of Technology	9.17337E+11
6	355959	India	Illinois Institute Of Technology	9.19827E+11
7	351520	India	Illinois Institute Of Technology	18369626042
8	372165	India	Illinois Institute Of Technology	9.19182E+11
9	369273	India	Illinois Institute Of Technology	9.16304E+11

Total rows: 15118

Query complete 00:00:00.125

```
23 SELECT * FROM CampaignData_Cleaned;
24
```

	id [PK] text	name text	category text	intake text	university text	status text	start_date date
1	AANF23	Gr Gs Fa24 Campaign- Admit, No Deposit	Post Admission	AY2024	Illinois Institute Of Technology	Completed	2024-03-20
2	AND23	Gr Gs Fa24 Campaign- Deposit No Action	Post Admission	AY2024	Illinois Institute Of Technology	Completed	2024-11-09
3	BPNAF...	Gr Gs Fa24 Campaign- Deposit, No I-20	Post Admission	AY2024	Illinois Institute Of Technology	Completed	2024-11-07
4	BPND23	Gr Gs Fa24 Campaign- In Progress	Pre Admission	AY2024	Illinois Institute Of Technology	Completed	2024-06-03
5	CTKANF23	Gr Gs Fa24 Campaign- Submit, Incomplete	Pre Admission	AY2024	Illinois Institute Of Technology	Completed	2024-08-03
6	DANE24	Gr Gs Call Campaign: India Anf	Pre Admission	AY2024	Illinois Institute Of Technology	Completed	2023-04-28
7	DNA24	Gr Gs Call Campaign: India No Deposit	Post Admission	AY2024	Illinois Institute Of Technology	Completed	2023-03-07
8	FA24AND	Gr Gs Call Campaign: Other Anf	Post Admission	AY2024	Illinois Institute Of Technology	Completed	2023-05-16
9	FA24DNA	Gr Gs Call Campaign: Other No Deposit	Post Admission	AY2024	Illinois Institute Of Technology	Completed	2023-05-07

Total rows: 23

Query complete 00:00:00.110

Here we have 2 select queries to ensure there are no duplicate records.
Result: (0 rows) (No Duplicates).

```
46 -- Duplicate Campaign IDs
47 SELECT id, COUNT(*)
48 FROM campaigndata_cleaned
49 GROUP BY id
50 HAVING COUNT(*) > 1;
51
52
```

Data Output Messages Notifications

	id [PK] text	count bigint
--	-----------------	-----------------

Total rows: 0 Query complete 00:00:00.099

```
46 -- Duplicate Campaign IDs
47 SELECT id, COUNT(*)
48 FROM campaigndata_cleaned
49 GROUP BY id
50 HAVING COUNT(*) > 1;
51
52
```

Data Output Messages Notifications

	id [PK] text	count bigint
--	-----------------	-----------------

Total rows: 0 Query complete 00:00:00.099

Here we have 2 select queries to discover any nulls or missing data and compare the result with the dashboard.

Result: (0 Nulls) .

```
51
52 SELECT
53     COUNT(*) FILTER (WHERE app_id IS NULL) AS null_app_id,
54     COUNT(*) FILTER (WHERE country IS NULL) AS null_first_name,
55     COUNT(*) FILTER (WHERE university IS NULL) AS null_email,
56     COUNT(*) FILTER (WHERE phone_number IS NULL) AS null_phone_number
57 FROM applicantdata_cleaned;
58
```

Data Output Messages Notifications

	null_app_id bigint	null_first_name bigint	null_email bigint	null_phone_number bigint
1	0	0	0	0

Total rows: 1 Query complete 00:00:00.088

```
59
60 SELECT
61     COUNT(*) FILTER (WHERE id IS NULL) AS null_id,
62     COUNT(*) FILTER (WHERE name IS NULL) AS null_name,
63     COUNT(*) FILTER (WHERE category IS NULL) AS null_category,
64     COUNT(*) FILTER (WHERE intake IS NULL) AS null_intake,
65     COUNT(*) FILTER (WHERE university IS NULL) AS null_university,
66     COUNT(*) FILTER (WHERE status IS NULL) AS null_status,
67     COUNT(*) FILTER (WHERE start_date IS NULL) AS null_start_date
68 FROM campaigndata_cleaned;
69
70
71
72
```

Data Output Messages Notifications

	null_id bigint	null_name bigint	null_category bigint	null_intake bigint	null_university bigint	null_status bigint	null_start_date bigint
1	0	0	0	0	0	0	0

Total rows: 1 Query complete 00:00:00.132

Here we have 2 queries to search more extensively for invalid data, which uncover: 3606 unknown values in the column country and 58 invalid values in the column Phone_number of the table Applicant_Data_Cleaned, We were already aware of this problem and to solve it, we should either remove these values so the hole rows or send a request to the company or client to retrieve these missing values which we can't in our case and to remove 3606 rows will definitely affect the results, so we choose to leave it marked as unknown.

```
96
97 SELECT DISTINCT country, count(*)
98 FROM applicantdata_cleaned
99 WHERE country ~ '[-]{2,}|[?]{2,}|N/A|Unknown|Invalid'
100 GROUP BY country|
101 ORDER BY country;
102
103
104
105
106
107
108
109
110
```

Data Output Messages Notifications



	country text	count bigint
1	Unknown	3606

```
89
90 --Uses ILIKE so it's case-insensitive (Invalid, invalid, INVALID all match).
91 SELECT
92     COUNT(*) FILTER (WHERE phone_number ILIKE 'Invalid') AS invalid_phone_number
93 FROM applicantdata_cleaned;
94
95
```

Data Output Messages Notifications



	invalid_phone_number bigint
1	58

We also have the same case in the column remark of the Outreach_Data_Cleaned table, and to make the invalid values clearer, we replaced them all with unknown, but also left them as they are for the same reasons.

```
53 SELECT
54     COUNT(*) FILTER (WHERE remark IS NULL OR remark = '') AS null_or_empty,
55     COUNT(*) FILTER (WHERE remark ILIKE 'Unknown') AS unknown_count,
56     COUNT(*) FILTER (WHERE remark ILIKE 'Invalid') AS invalid_count,
57     COUNT(*) FILTER (WHERE remark ILIKE 'N/A') AS na_count,
58     COUNT(*) FILTER (WHERE remark ~ '^+-$') AS dashes_count, -- one or more dashes
59     COUNT(*) FILTER (WHERE remark ~ '^[?]+$') AS question_marks, -- only question marks
60     COUNT(*) FILTER (WHERE remark !~ '[A-Za-z0-9 ,.\-]+$') AS other_symbols -- anything with odd symbols
61 FROM outreachdata_cleaned;
62
63
```

Data Output Messages Notifications

	null_or_empty bigint	unknown_count bigint	invalid_count bigint	na_count bigint	dashes_count bigint	question_marks bigint	other_symbols bigint
1	0	12913	1	0	17	0	89

```
118 SELECT
119     COUNT(*) FILTER (WHERE remark IS NULL OR remark = '') AS null_or_empty,
120     COUNT(*) FILTER (WHERE remark ILIKE 'Unknown') AS unknown_count,
121     COUNT(*) FILTER (WHERE remark ILIKE 'Invalid') AS invalid_count,
122     COUNT(*) FILTER (WHERE remark ILIKE 'N/A') AS na_count,
123     COUNT(*) FILTER (WHERE remark ~ '^+-$') AS dashes_count, -- one or more dashes
124     COUNT(*) FILTER (WHERE remark ~ '^[?]+$') AS question_marks, -- only question marks
125     COUNT(*) FILTER (WHERE remark !~ '[A-Za-z0-9 ,.\-]+$') AS other_symbols -- anything with odd symbols
126 FROM outreachdata_cleaned;
127
128 UPDATE outreachdata_cleaned
129 SET remark = 'Unknown'
130 WHERE remark ILIKE 'Invalid' -- catches 'Invalid' (any case)
131 OR remark ~ '^+-$' -- one or more dashes like --- or ----
132 OR remark ~ '^[?]+$' -- only question marks
133 OR remark !~ '[A-Za-z0-9 ,.\-]+$'; -- anything with odd symbols
134
```

Data Output Messages Notifications

	null_or_empty bigint	unknown_count bigint	invalid_count bigint	na_count bigint	dashes_count bigint	question_marks bigint	other_symbols bigint
1	0	13020	0	0	0	0	0

Here we have 2 queries, one that shows the Earliest_Start and Latest_Start of all campaigns and the second query shows Earliest_call_received and Latest_call_received, the problem here is that Earliest_call_received date is about 2 months before Earliest_Start date of campaigns, and in these 2 months, the calls are attached to campaigns that hasn't started yet, so my question here is for our supervisors who will grade this presentation and give us feedback, please tell me is this considered invalid data and should be removed or what?

```
109
110 select min(start_date) as Earliest_Start,
111        max(start_date) as Latest_Start
112 from campaigndata_cleaned;
113
114
```

Data Output Messages Notifications

	earliest_start date	latest_start date
1	2023-03-07	2024-11-09

```
114
115 select min(received_at) as Earliest_Receive,
116        max(received_at) as Latest_Receive
117 from outreachdata_cleaned;
118
119
```

Data Output Messages Notifications

	earliest_receive date	latest_receive date
1	2023-01-05	2024-12-20

Here we discovered that the outcome_1 column values can be renamed for better outcome and accuracy, so we replaced all values that have the meaning of connected to 'Connected' and all values that have the meaning of Not Connected to 'Not Connected'.

```
125 UPDATE outreachdata_cleaned
126 SET outcome_1 = 'Connected'
127 WHERE outcome_1 IN (
128     'Will Submit The Docx',
129     'Student Has The Needed Information, Does Not Need Assistance, And Plans To Enroll Soon',
130     'Will Start Application Soon',
131     'I901 Paid- Waiting For Slot',
132     'Application Already Stated',
133     'Ready To Pay The Deposit',
134     'Already Paid The Deposit',
135     'Want To Defer',
136     'Completed Application',
137     'Already Enrolled',
138     'Student Is Interested In Deferring To Fall 2025 (August)',
139     'Student Will Join Sp25 Session',
140     'Visa Approved- Travel Details Required'
141 );
142
143
144 SELECT outcome_1, COUNT(*) AS freq
145 FROM outreachdata_cleaned
146 GROUP BY outcome_1
147 ORDER BY freq DESC;
```

Data Output Messages Notifications

	outcome_1 text	freq bigint
1	Not Connected	10633
2	Connected	3680

Total rows: 19 Query complete 00:00:00.098

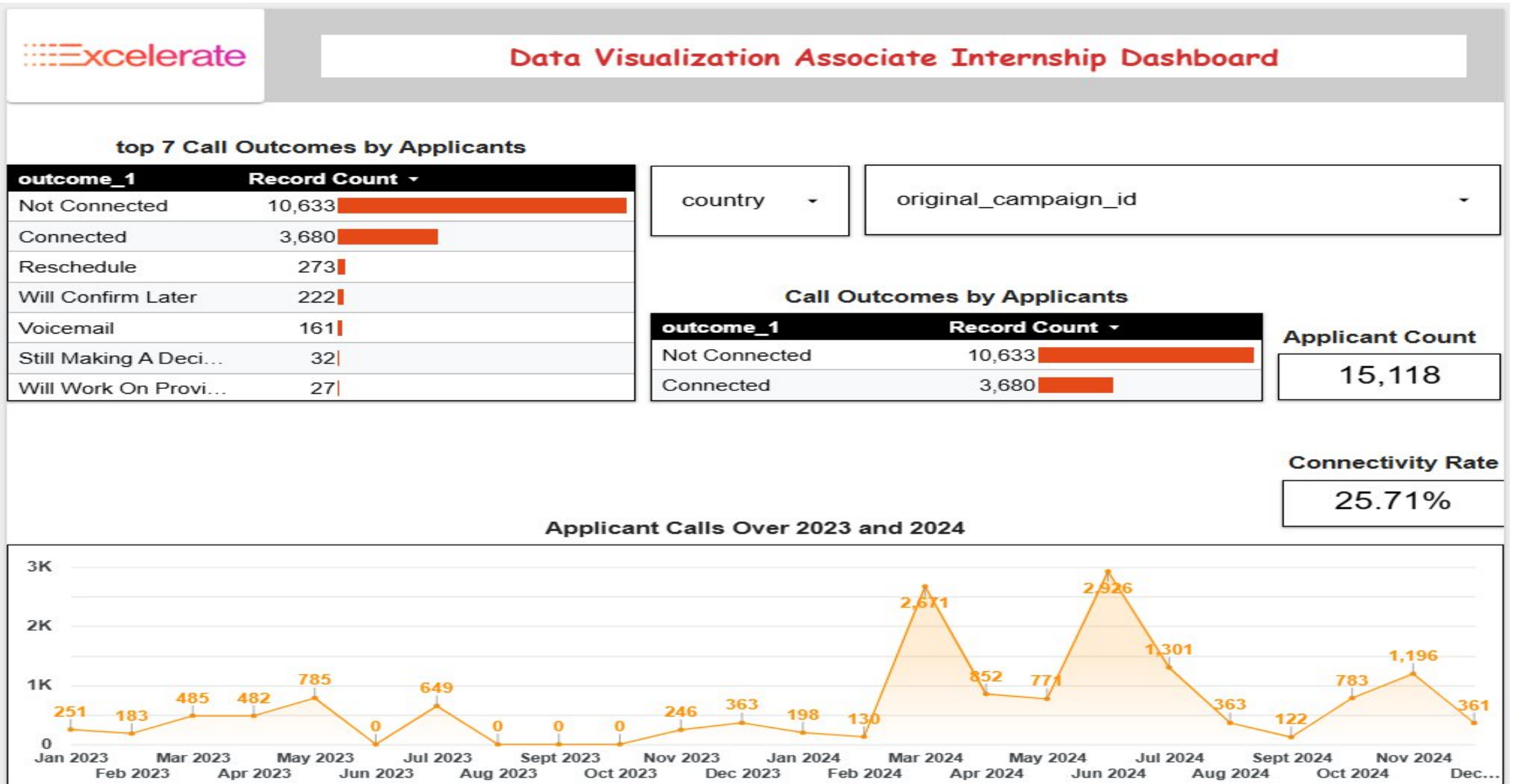
```
109
110 UPDATE outreachdata_cleaned
111 SET outcome_1 = 'Not Connected'
112 WHERE outcome_1 IN (
113     'Disconnected',
114     'Wrong Number',
115     'Not Interested To Iit',
116     'Not Interested',
117     'Student Will Not Be Attending Illinois Tech And Needs To Be Withdrawn',
118     'Not Interested To Pay',
119     'Student'S Visa Was Denied,They Are not Interested In A Deferral, And They Would Like A Refund Of The Er
120 );
121
122
123
124
```

Data Output Messages Notifications

	outcome_1 text	freq bigint
1	Not Connected	10633
2	Connected	3680
3	Reschedule	273
4	Will Confirm Later	222
5	Voicemail	161
6	Still Making A Decision	32
7	Will Work On Providing Documents Soon, Still Interested In Fa24	27

Total rows: 19 Query complete 00:00:00.098

Dashboard Walkthrough:



Dashboard Walkthrough:

4-page interactive dashboard built in Looker Studio.

- **The first Page (Applicants Overview):**

- It highlights a low overall connectivity rate (25.7%), with most calls ending as Not Connected.
- Applicant Count is 15,118 and Our time frame for is from 05-01-2023 to 20-12-2024, and in that time the highest number of Applicants were in 2024 (March by 2,671 calls),(June 2,926 calls), (Juley by 1,301 calls), and (November by 1,196 calls), on the other hand the least numbers were in 2023 (June, August, September, October) all by (0 calls).
- We also have two drill-down lists (Country and Campaign_name) to allow users to interact with the data and customize their view.

- **The second page (Countries Performance):**

- It displays the highest countries that has applicants, Top 5 countries that has applicants whose Outcome Call is 'Connected' are (India, Unknown, Ghana, Nigeria, and South Africa, on the other hand, there are a lot of countries with 0 applicants.

country ▾

original_campaign_id ▾

Campaign Count

23

Applicant Count

15,118

top 7 Countries by Applicants

country	app_id ▾
India	5,287
Unknown	3,606
Ghana	1,729
Nigeria	1,315
Pakistan	565
South Africa	378
Bangladesh	246

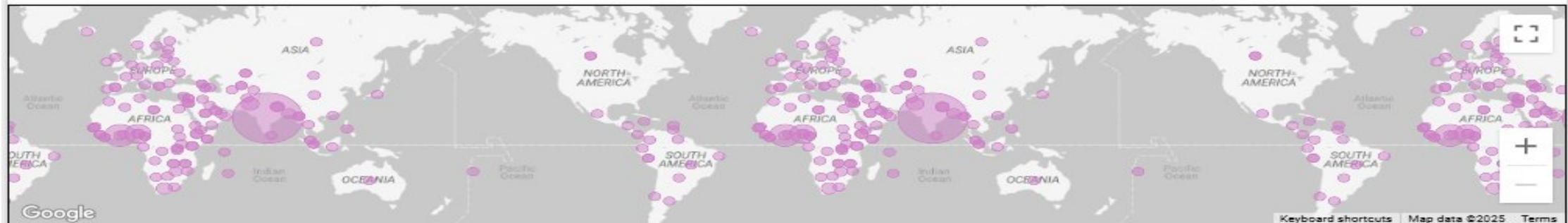
Top 7 Countries by Connected Applic...

country	Connected ▾
India	1,254
Unknown	916
Ghana	530
Nigeria	387
South Africa	149
Pakistan	93
Bangladesh	53

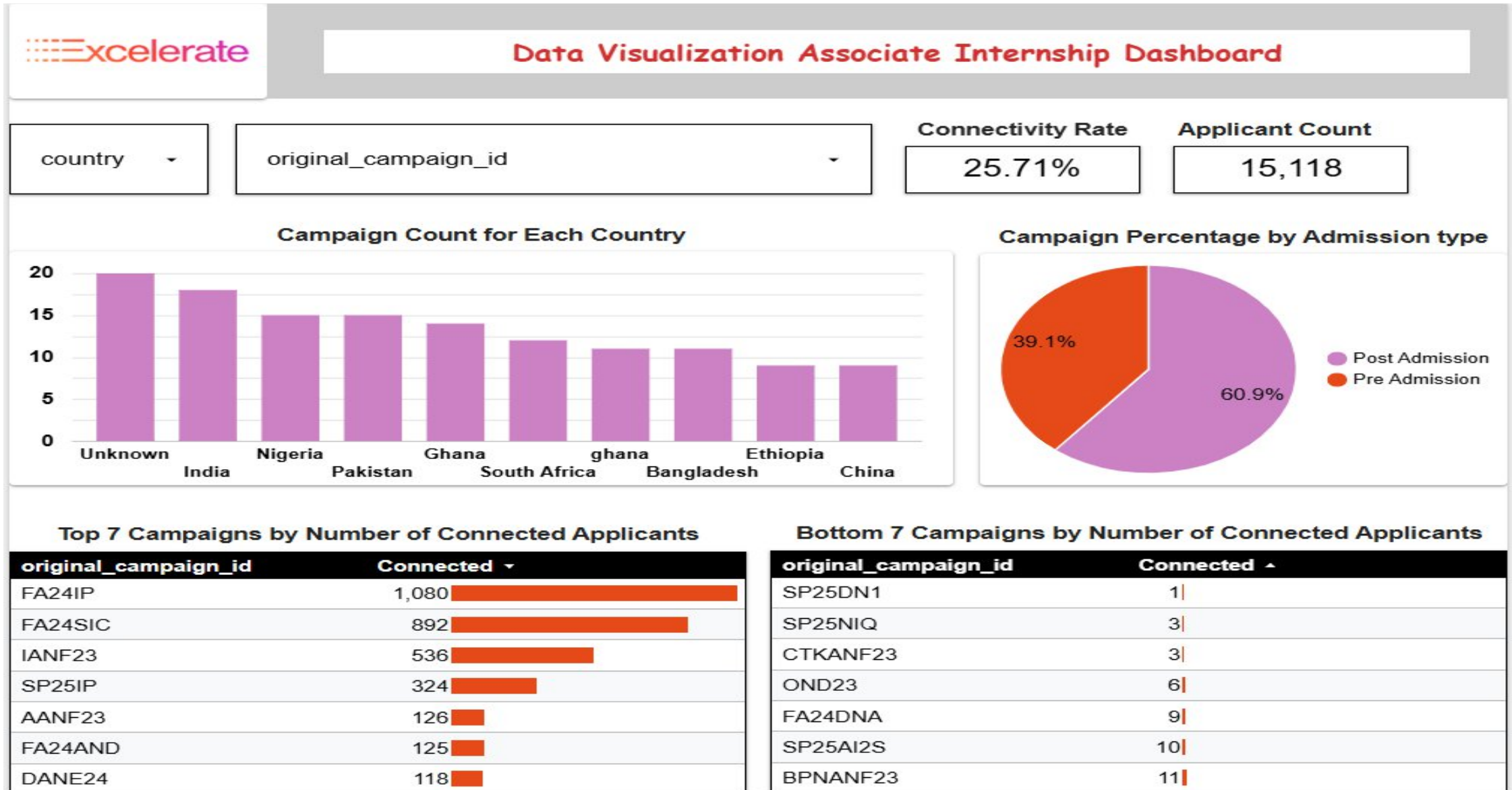
Bottom 7 Countries by Applicants

country	app_id ▲
Ireland	1
armenia	1
grenada	1
Belarus	1
Cyprus	1
Latvia	1
kazakhstan	1

Number of Applicants by Country



The third page (Campaigns Performance):



- **The third page (Campaigns Performance):**

- Which shows us the highest and lowest countries for campaign number and how many for each, as we can see India is dominant by 18, and China is the weakest by 9.
- a pie chart which displays the percentage of post- admission campaigns with 60.9% vs pre-admission campaigns with 39.1%.
- The top 5 campaigns are (FA24IP : 1,080 Applicant), (FA24SIC : 892 Applicant), (IANF23 : 536 Applicant), (SP25IP : 324 Applicant), and (AANF23 : 126 Applicant), on the other hand bottom 5 campaigns are (FA24DNA : 9 Applicant), (OND23 : 6 Applicant), (CTKANF23 : 3 Applicant), (SP25NIQ : 3 Applicant), and (SP25DN1 : 1 Applicant).

- **The Fourth and last page (Agents Performance):**

- Which shows us that Rudra handled 4,840 calls over 2023, 2024 and Prajwal 3,762 calls, the highest 2 by far, on the other hand, Twinkle handled 26 calls over 2023, 2024, Namrata 110 calls, and Shrutish 309 which is very low.

- **[Looker Studio Dashboard Link](#)**

country ▾

original_campaign_id ▾

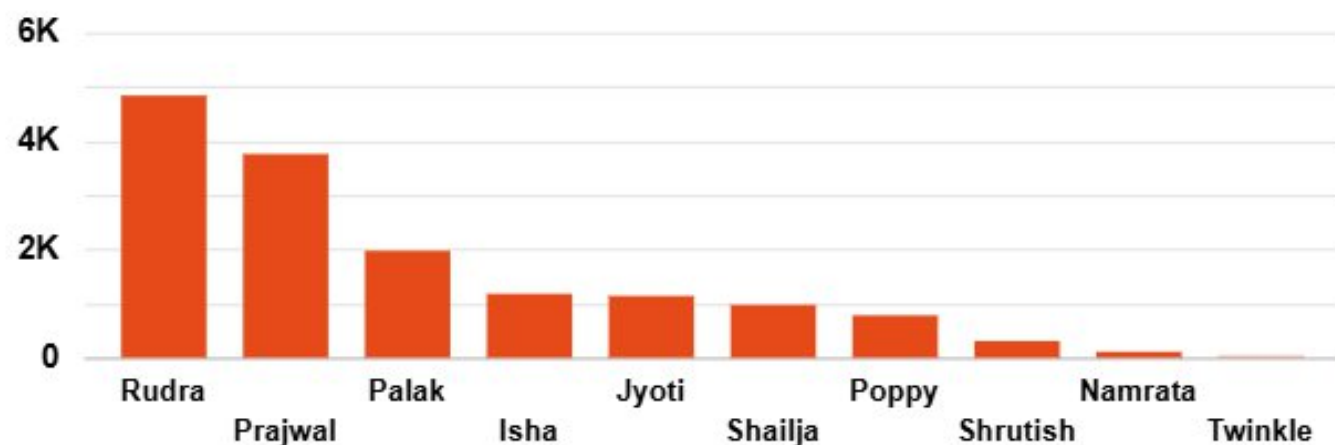
Connectivity Rate

25.71%

Applicant Count

15,118

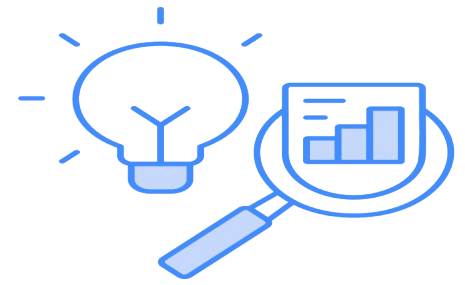
Number of Calls by Agent



Top 7 Agents by Connected Calls

caller_name	Connected ▾
Prajwal	1,088
Rudra	1,080
Palak	377
Isha	275
Shailja	254
Jyoti	240
Poppy	225

Key Findings

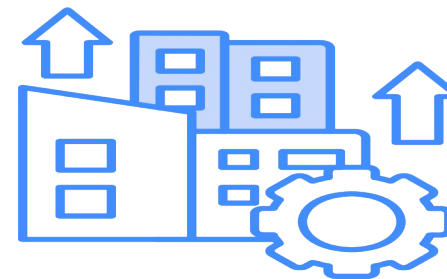


1. High Data Consistency – All tables have uniform data types and formats, ensuring accurate joins and reliable analysis.
2. 'Unknown' Values – 3,606 unknown countries and 58 invalid phone numbers limit geographic and contact-based insights.
3. Early Call Activity – Calls recorded up to two months before campaigns hint at pre-launch engagement or date-entry errors.
4. The highest responses from applicants to the agents' calls, was (not connected), and the Connectivity Rate is so low (25.71%) which certainly triggers a red flag.
5. The highest 5 countries that has applicants in other words calls are India, Ghana, Unknown, Pakistan, and Nigeria, so that tell us how much is our success in these countries, on the other hand, we have a lot of countries that has only 1 or 2 applicants like Irland, Armenia, Grenada.

Key Findings

6. Top 5 countries that has applicants whose Outcome Call is 'Connected' are (India, Unknown, Ghana, Nigeria, and South Africa, on the other hand, there are a lot of countries with 0 applicants.
7. Our time frame for calls is from 05-01-2023 to 20-12-2024, and in that time the highest number of calls were in 2024 (March by 2,671 calls),(June 2,926 calls), (Juley by 1,301 calls), and (November by 1,196 calls), on the other hand the least numbers were in 2023 (June, August, September, October) all by (0 calls).
8. Rudra handled 4,840 calls over 2023, 2024 and Prajwal 3,762 calls, the highest 2 by far, on the other hand, Twinkle handled 26 calls over 2023, 2024, Namrata 110 calls, and Shrutish 309 which is very low.
9. The top 5 campaigns are (FA24IP : 1,080 Applicant), (FA24SIC : 892 Applicant), (IANF23 : 536 Applicant), (SP25IP : 324 Applicant), and (AANF23 : 126 Applicant), on the other hand bottom 5 campaigns are (FA24DNA : 9 Applicant), (OND23 : 6 Applicant), (CTKANF23 : 3 Applicant), (SP25NIQ : 3 Applicant), and (SP25DN1 : 1 Applicant).

Recommendations



To address the findings and strengthen future operations, we propose:

1. Data Quality & System Hygiene

- **Data Enrichment Efforts**

- Collaborate with the client/company to retrieve missing country and phone number details where possible.
- If retrieval isn't feasible, clearly flag these unknowns in all visualizations to maintain transparency.

- **Investigate Pre-Launch Calls**

- Conduct a deeper audit to confirm whether early call records reflect actual outreach or are errors.
- If they are valid, consider leveraging this early engagement as part of future campaigns.

- **Standardized Data Entry Protocols**

- Create and enforce a strict data-entry procedure to prevent future “Unknown” values and inconsistent date entries.

- **Regular Data Audits**

- Schedule routine consistency checks to ensure ongoing data quality and early detection of anomalies.

2. Improve Applicant Connectivity

- **Problem:**
 - Connectivity rate is critically low (25.71%).
 - “Not Connected” is the most common outcome.
- **Recommendations:**
 - **Agent Training & Scripts:** Revise outreach scripts and provide targeted training to improve engagement.
 - **Call at the right time:** Align agent schedules with applicant local time zones.
 - **Follow-Up Protocols:** Implement structured follow-up sequences for unresponsive applicants.
 - **Channel Diversification:** Introduce WhatsApp, email, or SMS outreach to supplement calls.
 - **Why it matters:** Even a 5–10% lift in connectivity means thousands more engaged applicants.
- **SMS pre-notification:** send 1 SMS 24 hours before scheduled call (template A/B test).
- **Two-call strategy:** 1 initial + 1 follow-up at different times/days before marking Not Connected.

3. 🌐 Country Strategy

- **Focus on top markets:** Double down on India, Ghana, Pakistan, Nigeria (optimize campaigns and staffing here).
- create country-specific playbooks (local language scripts, ideal call windows, SMS content).
- **Competitor benchmarking:** In low-volume countries, analyze competitor companies' marketing (channels, timing, offers) to identify gaps.

4. 🧑💼 Agent Performance

- **Immediate actions:**
 - Validate agent attribution accuracy (ensure calls attributed correctly to agents like Rudra, Twinkle).
 - Rebalance workloads: reduce spare/underused agents; reassign top performers for high-value campaigns.
 - **Reward quality, not just volume:** Incentivize connected calls that lead to deposits/enrollments.

4. Agent Performance

- **Performance management:**
 - For top performers (e.g., Rudra): implement shadow sessions to extract best practices, share their techniques with the team.
 - For low performers (e.g., Twinkle): coaching session with call-review, 30-day improvement plan, and clear KPIs (Connected rate, calls/hour, quality score).
- **Incentives:**
 - Introduce short-term incentives tied to Connected conversion and quality (not just call count).
- **Reporting:**
 - Weekly agent dashboard: calls, connected rate, avg talk time, conversion to next funnel step.

5. Campaign-level recommendations

- **Prioritize high-performing campaigns for scale:**
 - Expand outreach budget and channel mix for FA24IP, FA24SIC, IANF23, SP25IP (top 5 by applicants).
 - Replicate messaging/targeting for similar geographies/demographics where these campaigns succeeded.

5. Campaign-level recommendations

- **Rescue low-performing campaigns:**
 - Audit creative, offer, and target lists for FA24DNA, OND23, CTKANF23, SP25NIQ, SP25DN1; run small experiments (re-targeting, different call scripts, localized creatives), or sunset them.
- **ROI tracking:** Add a simple cost-per-connected-applicant metric to compare campaigns fairly, if spend data available.
-  **Timing & Seasonality**
 - **Peak staffing:** Staff up in March, June, July, and November (historically highest call months).
 - **Zero-call months:** Investigate why June–Oct 2023 had no calls; set alerts to prevent future gaps.
 - **Channel layering:** If no answer after 2 calls, switch to WhatsApp/SMS before final attempts.
 - **Seasonal campaigns:** Align campaign launches with academic calendars (application deadlines, exam results, visa cycles).

6. Experiments and tests (to validate improvements)

- **A/B test 1:** SMS pre-notice vs no SMS — measure Connected rate lift after 2 weeks.
- **A/B test 2:** Two-call strategy vs single-call — measure incremental Connected rate and cost per connected.
- **A/B test 3:** Call-time slots (morning vs evening local time) for top 5 countries to find optimal windows.
- **Track outcome:** run experiments with clear sample size and stop rules; report lift and statistical significance.

The End